

MERAKI LABS · VENTURE STUDIO

Full Stack Engineer — Work Trial

Kindly pick either one of the following problem statements

2–3 days · GitHub + Proposal Doc + AI Build Log + Live Presentation Pick ONE problem from the set below.

What This Is

We are evaluating how you *think*, how you *build with AI*, and how you *communicate* under realistic constraints — not how much you can ship. The product described in each problem is a **north star, not the assignment**. You are designing the whole system on paper and building one real slice of it. Do not spend more than 2 focused days building. Scope is the test.

Guiding Principles

- **Show your thinking and your AI workflow, not just code.** The proposal doc and your prompt artifacts matter as much as the repo.
- **Systems design first.** We want the whole-system blueprint even though you'll build only one module.
- **Production-first.** What breaks at 100k+ users across multiple tenant startups? What would you monitor? What would you fix first?
- **Build with AI, transparently.** We want your *real* prompts, planning docs, and rules files — including where the AI failed you and how you recovered.
- **Scope ruthlessly.** Tell us what you chose *not* to build and why. That is part of the answer.
- **Be honest about limitations.** Tell us what's still unbuilt, not just where it works.

Problem Statements

PS 1 — Venture-Studio CRM (multi-tenant growth engine) *North star:* a CRM serving multiple startups under one studio, cross-utilized for performance-marketing tracking, lead generation, lead-cycle management, AI-assisted automated outbound + reply handling, and sales-cycle management.

Design the whole system; build **one** working sub-module — pick the highest-value slice, e.g.:

- *AI-assisted outbound:* generate and personalize a sequence from a lead record, with human-in-the-loop approval and automated reply classification.
- *Lead-cycle state machine:* lead → MQL → SQL → won/lost, with event tracking and a clean API/UI.
- *Performance-marketing attribution:* ingest campaign + lead data and attribute pipeline back to source.

The **multi-tenant isolation** decision is the crux — how do many startups share one platform without leaking each other's data? Address it explicitly.

PS 2 — Startup Hiring System (application → AI screen → rubric → decision) *North star*: application submission, AI video screening, rubric-based grading and selection, and application tracking. The admin panel uploads a JD and **auto-generates rubric criteria** to drive evaluation.

⚠ *Scope guardrail: do not process real biometric or video data.* Mock or stub the video input (a transcript or sample file is fine). We care about the design, the JD→rubric generation, and the grading logic. If you engage with **fairness/bias and compliance** (adverse-impact checks, NYC LL144-style auditing, EU AI Act framing), that's a strong senior signal.

Design the whole system; build **one** working sub-module — e.g.:

- *JD → rubric generator*: upload a JD, auto-produce weighted, defensible, auditable evaluation criteria.
- *Rubric grader*: given a candidate artifact + rubric, produce an evidence-cited score with safeguards against hallucinated grades.
- *Applicant tracker*: ATS-style state machine + admin panel.

Required Deliverables (both problems)

1. **Systems Design Doc** — architecture diagram, data model, key trade-offs, what scales and what breaks, and the crux addressed head-on (multi-tenant isolation for PS1; fairness/auditability for PS2).
2. **AI Build Log** — *this is what we're really testing*. Submit the actual artifacts of how you drove the AI IDE:
 - The planning doc / PRD / spec you fed the tool.
 - Your rules / context files (`.cursorsrules`, `CLAUDE.md`, agent or system prompts).
 - The real prompts you used to plan, build, and debug — **including the ones that failed and how you course-corrected**. Show the messy middle, not a highlight reel.
 - Which IDE / model / tools, and why.
3. **Working sub-module** — runnable from a clean README. If we can't run it, that's part of the evaluation.
4. **Proposal Doc** — like a founding-engineer proposal: what you'd build, in what order, what you deliberately deferred, and what you'd tackle with two more weeks or at 100k users.

Submission

1. **GitHub repo**. Clean README. We should be able to run the module.
2. **Proposal Doc + AI Build Log**. PDF or Notion; the build log can live in `/ai` in the repo.

3. **Live presentation. 20–30 minutes.** Walk us through the design, the module, and your AI workflow. We push hardest here — be ready to defend trade-offs and to explain a prompt that *didn't* work.

What We Are Not Looking For

- A finished product. Two days is not enough and we know it.
- Over-engineering. A six-service architecture for one slice is a red flag.
- A sanitized prompt transcript that hides the real iteration.
- Code — including AI-generated code — that you cannot explain.

Questions before you start? Reach out — questions are a good sign. **Recruiter:** Ritesh Kalvelli — ritesh@merakilabs.com